

ATTIC ACCESSES: HIGH PRIORITY OR NO BIG DEAL?

Some calculations answer the question, How much does proper attic access insulation matter?

BY JIM MELESKY

My firm, Energy Sentry Solutions, provides a product—the Energy Guardian—to insulate and seal attic accesses. In the course of everyday business, I have come upon a recurring difference of opinion among weatherization professionals regarding the importance of attic upgrades. Indeed, all weatherization programs address attic accesses with a wide variety of remediation measures. How much energy and money will the crew save the home's occupants by insulating and sealing the attic access? The answer can have a huge impact on projected savings, on the amount budgeted for this improvement, and on the type of measures used to upgrade this 5–10 ft² area of the home.

How Much Do Attic Accesses Matter?

To evaluate the consequences of different remediation measures, let's first examine the impact of attic pull-down ladders and pop-up hatches on the entire attic thermal barrier. While the measurements of the opening vary slightly, a typical attic ladder creates a 10 ft² opening. The ladder is attached to 1/4-inch plywood that has an R-value of 0.31.

There are some who insist that the lack of insulation over an attic access is of little consequence; they calculate the impact of not insulating the access by using an average R-value formula. In this case, we will examine a home with a



Alexis Garcia pokes her head through an attic access.

1,000 ft² attic that has R-38 insulation in the attic floor. The attic access is a pull-down ladder. To calculate the average R-value of the attic they use this formula:

$$[(38 \times 990 \text{ ft}^2) + (0.31 \times 10 \text{ ft}^2)] / 1,000 \text{ ft}^2 = R-37.6$$

If this is correct, the pull-down ladder reduces the thermal barrier for the entire attic from R-38 to R-37.6. It would follow that the upgrade does not warrant much emphasis. In fact, this analysis is incorrect, but unfortunately people continue to believe in it to this day.

The DOE Weatherization Assistance program's *Residential Energy Workbook* explains in plain language how energy is

wasted in existing homes and offers a host of ways to upgrade homes for cost savings and improved comfort. The workbook addresses this specific matter of attic hatches with the following formula:

A/R for the pull-down + A/R for the rest of the attic = A/R total, where A = area and R = R-value.

$$990 \text{ ft}^2 / 38 + 10 \text{ ft}^2 / 0.31 = 58.31$$

$$U = UA/A, \text{ where } U = U\text{-value}$$

$$58.31 / 1,000 \text{ ft}^2 = 0.05831$$

$$\text{Effective R-value} = 1/U = 17.1$$

This formula shows that the impact of not insulating an attic access is dramatic. The pull-down ladder reduces the R-value for the entire attic by 55%.

That means that the effective R-value for the attic is only R-17, although some people would say that it is R-38. Even a pop-up hatch has a big effect on the entire attic R-value. If we apply the same formula to a 22 1/2 inch x 26 inch hatch opening in the same attic, the effective insulating value is reduced to R-24 for the entire attic.

What about air leaks? Even if rolled or blown-in insulation covers a hatch, air can flow through this insulation. As a consequence, energy will still be lost through the attic access due to the air leak.

In addition, pull-down ladders have air leaks on all four sides of the plywood housing the ladder and almost always in the four corners of the wood framing. The two short sides of the ladder unit contain the widest gaps. There is approximately a 1/2-inch gap along the side where the ladder is hinged to the plywood. Opposite the hinged side is another 1/2-inch gap that gets larger every year—with the passage of time, the plywood will warp from the ladder being pulled down. It is not uncommon for this gap to grow to 1 inch. The framing for the ladder is usually made of 1 inch x 4 inch boards. Each corner will have a 1/4-1/8-inch gap that is 4 inches long. Gaps of various sizes also exist between the flooring and the framing for the pull-down ladder. At the low end, these gaps around pull down ladders add up to 40 square inches or more of leakage area. While the pop-up hatches come in various sizes and shapes, they pose the same basic problem, for they have significant gaps as well.

Anyone who doubts the effect of these air leaks need only stand near the access on a windy day or take a blower door test and follow the wind tunnel. If the house is reasonably sealed, the wind tunnel will lead directly to the attic pull-down ladder or hatch.

Is this just a winter problem? During the winter, hot air in the home rises and escapes through the attic access. Hot air is less dense than cold air and will naturally rise. This, along with the normal air flow in the home, creates the stack effect. It is similar to the way hot air moves upward in a chimney.

In the summer, the opposite occurs as hot air from the attic moves into the living area. It does this for several reasons. In principle, hot air will be drawn from the attic into the home by a fan or air conditioning unit in the home and by any breeze flowing through the attic via the vents. The air flow in the home will move any radiant heat built up on the hatch or pull-down cover into the living area. It will also move any convective heat that flows through the very low thermal barrier. Finally, as air



Once a chimney is sealed, it is safe to assume that it will not be disturbed. That is not the case with attic ladders and hatches. Is there an unsealed attic access under this insulation? If so, air leaks could be a big problem.

moves in the attic, the hot air will flow through air leaks in the attic access much as it does through the attic vents.

One must also take into account the cost of heating a home versus the cost of cooling it to determine summer cost versus winter cost. While precise comparison requires far more detailed information than this article allows, it is clear that considerable energy is lost in the summer. In a practical sense, then, the homeowner pays to heat the attic during the winter and cool it during summer.

Quantifying the Impact

To determine how much air leaks and degrading thermal barriers affect a home, let's consider the case of a two-story house with nearly 2,500 ft² of floor area that is located in Chicago. A weatherization crew has successfully accomplished a number of upgrades within budget, such as adding insulation

to the attic; air sealing the attic, including the attic hatch and the chimney; and installing a blanket on the water heater. The blower door test shows positive results, with a pretest reading of 4,931 CFM₅₀ and a posttest reading of 3,501 CFM₅₀. The weatherization crew followed agency procedure concerning the priority of upgrades. While a number of valid measures were accomplished, the crew focused on sealing and insulating the attic.

Applying the savings-to-investment ratio (SIR) that weatherization professionals understand, a cost-effective upgrade produces a result of 1 or greater.

In calculating the SIR, we use the effective thermal results and air leakage given above. We will assume that the home has oil heat. We will use conservative costs for fuel: \$1.20 per gallon for oil and \$0.10/kWh for electricity. If \$250 is spent to upgrade the pull-down ladder, the SIR for this measure is 6.28. If this same home has a pop-up hatch and \$150 is spent to insulate and seal that hatch, the SIR is 3.71.

When we calculate the SIR, we see that sealing and insulating attic accesses is very cost-effective. However, the type of upgrade implemented by weatherization agencies can vary widely.

Some agencies use measures that ensure a long-term, durable solution. One such measure is to install a hinged plywood container in a reverse u shape, filled with high R-value insulation. Another is to place a box-shaped plywood structure with high R-value insulation over the opening in the attic. Both measures have heavy-duty sealing and are secured by a hook and eye or strapping. These measures are very durable, but they require more time to make and a higher degree of expertise than simpler, but less effective measures.

Others agencies are not concerned about the longevity of the solution. If the upgrade passes an inspection shortly after it is completed, they think it is good enough. A typical example of this short-term approach for attic ladders is to use insulation board to create a box over the opening. The box is sealed with weatherstripping.

For hatches, insulation can be glued to the top of the hatch cover and weatherstripping applied to the perimeter of the hatch cover. While these measures cannot withstand wear and tear, they will produce blower door test results similar to the results achieved by more substantial and long-lasting upgrades. These less stringent alternatives cost less, but they only appear to have a sufficient SIR.

Does durability affect energy saving? The answer would seem to be yes. However, neither budgets nor procedures for weatherization programs allow for postinspections throughout the period of time that energy must be saved in order to justify an upgrade. Such inspections would reveal whether or not the upgrade remains intact and effective.

Nonetheless, since durability is a key component in forecasting energy savings relating to each upgrade, we must take a sober look at the realistic life of each upgrade if we want to make accurate projections.

The SIR calculation used by many weatherization agencies specifies an assumed period of time that a particular upgrade will remain effective. For an attic access, the upgrade is usually assumed to last at least 13 years.

Durability standards for various attic upgrades are necessarily more stringent in some cases than in others. Take the sealing around a chimney and around an attic access as examples. Both are very important, and both must last for the same period of time. However, once a chimney is sealed, it is safe to assume that it will not be disturbed. That is not the case with attic ladders and hatches. By building code, they provide a required means to gain access to the attic. That means that the upgrade will be subject to repeated use



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and to impact with other objects—such as flooring, floor joists, or rafters—as well as with anything stored in the attic, each time the access is opened. This will wear both the air seal and the insulation. If the ladder and hatch upgrades are not made to withstand wear and tear, the SIR will be significantly reduced.

Returning to our Chicago home again, let us analyze the low-cost and less durable measures used to insulate and seal the attic accesses. For these upgrades, many agencies allocate \$150 for the pull-down ladder and \$80 for the hatch. Again, these upgrades may provide appropriate insulation and sealing of both attic accesses, but they are not usually durable. If through normal wear and tear the upgrades last only one year, the SIR falls to 0.80 and 0.53 respectively for the pull-down ladder and the hatch.

When they fall below the 1 thresh-

old, these upgrades are no longer cost effective. This is true even though they cost less to install, and although they could pass an initial weatherization inspection.

It is clear that durability matters.

Can the durability of the upgrade for attic accesses affect other valid upgrades?

Returning to our example home, let us change the circumstances slightly. In this case, the home has a pull-down ladder and only R-19 insulation in the attic. An additional R-19 is blown into the attic at a cost of \$850 (\$0.85/ft²). The pull-down ladder is also insulated and sealed, using the less durable \$150 solution. Both upgrades seem to be cost-effective, as they produce an SIR in excess of 1. When the pull-down upgrade fails after one year, as we already know, it becomes an ineffective measure. However, it also has a devastating impact on the insulation that was blown into the attic. The R-38 insulation is reduced to an effective value of R-17. The added R-

19 insulation is reduced to an effective value of R-8.5. From a value point of view, that more than doubles the cost of the effective R-value/ft².

In short, \$1,000 was invested in a home that will not perform as intended. This example shows that the two upgrades cannot be viewed as separate and distinct measures.

What Should the Standards Be?

It comes as no surprise that there is a move to tighten up and improve the standards. The people behind this move have strong opinions. John Ohm, president of Ohm Weatherization in Bethlehem, Pennsylvania, stresses the importance of the attic accesses. "For a long time this upgrade has been undervalued for its budget and importance. We have to accept the fact that home-

owners will go into their attic whether they have a hatch or attic ladder. That is why the upgrade must be made to withstand a lot of wear and tear. As an industry, we must emphasize durability and make it part of the standards. It costs a little more, but the job must be done right for the homeowner to get the expected energy savings.”

Vic Alshire, who is the president of Comfort Company in Washington Court House, Ohio, puts it succinctly. “It’s really quite simple. Kits are far more cost-effective, but only if they meet all the important standards. You need high R-value, a tight air seal, and long-term durability. In addition, attic ladders and hatches must have protective barriers to keep insulation from entering the living area. The upgrade must have a two-piece design that creates a barrier while also providing access to the attic. Therefore, any single-piece unit is fatally flawed for use in most homes. Any upgrades for attic openings short of all these requirements are just window dressing and a waste of money and time.”

“We understood the problem some time ago,” says Kevin Soucy, president of Affordable Energy Solutions LLC, in Milford, New Hampshire. “The flimsy solutions used in the industry for years just don’t cut it. The upgrades must last as long as the savings are projected. While community action agencies and utility programs are very cost conscious, they still want the projected cost savings to be real. But making an acceptable product with our crews takes a long time.”

Existing practice and inertia can be powerful forces to resist change. Fortunately, science clarifies the matter for us. It is instructive regarding the effective R-value calculations, air leakage, SIR calculations, and the interdependency of upgrades. The conclusion is very straightforward.

First, attic ladders and hatches are extremely important. They cause energy loss in both summer and winter. Second, careful scrutiny is needed to ensure that the upgrade for an attic access has the proper design, providing a protective barrier. Third, because the attic access affects other upgrades in a

very significant way, it must conform to higher durability standards than other seemingly similar measures.

In short, upgrades must be durable enough to last 12 or more years, must create a barrier to keep insulation from entering the home, must have an R-value similar to the rest of the attic insulation, and must provide an air seal.

Funding for the upgrades needs to reflect the work and the products necessary to meet more stringent standards. It is increasingly common for agencies to allocate \$175–\$225 for hatches and \$250–\$300 for pull-down ladders. Long-term durability is best achieved through warranted work or products and a commonsense review of the standard that takes wear and tear into account.

A word about the future: Free enterprise has always set the stage to replace individually made products with equal or better ones that can be mass produced. As the contest for better and more cost-effective home energy saving solutions continues, the ultimate winner will be the homeowner and the environment. That’s a very good thing to give the next generation.



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Several attic access kits, including the Energy Guardian, are available that provide a less labor-intensive alternative to individually made upgrades. These kits range in price from \$75 to \$180. Shipping may cost extra, depending on the vendor. The kits vary widely in terms of durability, R-value, design, and ease of installation.